

HW5

Greyson Knippel
MATH 339 — Gonzaga University
compiled with L^AT_EX

October 10, 2026

Problem 19

(a)

$$\mathbf{a}_1 = \begin{bmatrix} -1 \\ 3 \\ 2 \end{bmatrix} \quad \mathbf{a}_2 = \begin{bmatrix} 2 \\ -6 \\ -4 \end{bmatrix}$$

$$\mathbf{a}_2 = -2\mathbf{a}_1 \implies 2\mathbf{a}_1 + \mathbf{a}_2 + 0\mathbf{a}_3 = \mathbf{0}$$

(b)

$$\mathbf{a}_1 + \mathbf{a}_2 = \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix} = \mathbf{a}_3$$

$$\mathbf{a}_1 + \mathbf{a}_2 - \mathbf{a}_3 = \mathbf{0}$$

Problem 37

$$c_1\mathbf{w}_1 + c_2\mathbf{w}_2 + c_3\mathbf{w}_3 = \mathbf{0}$$

$$c_1(\mathbf{v}_1 + \mathbf{v}_2) + c_2(\mathbf{v}_2 - \mathbf{v}_3) + c_3(\mathbf{v}_2 + \mathbf{v}_3) = \mathbf{0}$$

$$c_1\mathbf{v}_1 + (c_1 + c_2 + c_3)\mathbf{v}_2 + (-c_2 + c_3)\mathbf{v}_3 = \mathbf{0}$$

S linearly independent $\implies$

$$c_1 = 0$$

$$c_1 + c_2 + c_3 = 0$$

$$-c_2 + c_3 = 0$$

$$c_3 = c_2 \implies 2c_2 = 0 \implies c_1 = c_2 = c_3 = 0$$

soT is linearly independent.